1 Question (20 points)

The divine ratio of four numbers is calculated as

$$H = \frac{\frac{1}{p} - \frac{1}{q}}{\frac{1}{r} - \frac{1}{s}}.$$

Complete the following program which **calculates** and **prints** the divine ratio of the four integers p, q, r, and s. You may declare any necessary variables. You should not change any part of the given code.

```
public .......  Divine{
    public ....... ....... main(String[] args){

        int p=14, q=23, r=8, s=13;

        .
        .
        .
        .

        System.out.println("Divine ratio is equal to " + h);


    }
}
```

2 Question (15 points)

What will the output of the following program?

```
int sum = 0;
for (int i = 1; i <= 5; i++){
  for (int j = i; j <= 5; j++){
    for (int k = j; k <= 5; k++){
      sum = sum + 1;
    }
  }
  System.out.println(sum);
}
```

3 Question (15 points)

In each part, write a for loop to produce the given output.

```
a) -6 6 18 30 42
b) 1 27 125 343 729
c) 1 4 9 16 25
```

4 Question (20 points)

Write a structured and non-redundant program (including class and main method declaration) that will print the following figure. (You do not need to use for loops.)

```
\\""//
||||||
//""\\
||||||
\\""//

||||||
\\  //
//""\\
\\  //
||||||

//""\\
||||||
\\  //
//""\\
\\  //
```

5 Question (15 points)

The maximum distance that a car can travel depends on its fuel consumption, the number of passengers in the car and the number of liters of fuel in the fuel tank. Suppose that there is a car with 48 liters of fuel, 5 liters per 100 km fuel consumption (it consumes 5 liters of fuel in every 100 km with the driver inside only)) and 4 passengers. Every additional passenger increases the fuel consumption by %5, i.e, 3 additional passengers increases the fuel consumption by %15.

Write a method to calculate the maximum distance that can be traveled by the car and print it.

Define at least 4 variables in your solution.

6 Question (15 points)

Assume that there are 30 days in a month and 365 days in a year. You are given an integer variable d that holds the total number of days. Write a program to convert d into days, month and year and print it. For example, 10000 days is equal to 27 years, 4 months and 25 days.